Terraform apply:

```
joaoribeiro@JOCAPC:~/remote-state-example/terraform$ terraform apply -auto-approve
module.payroll_europe.aws_dynamodb_table.payroll_db: Refreshing state... [id=PayrollDB-EU-joao]
module.payroll.aws_dynamodb_table.payroll_db: Refreshing state... [id=PayrollDB-joao]
module.payroll.aws_s3_bucket.payroll_docs: Refreshing state... [id=payroll-docs-us-east-1-joao]
module.payroll_europe.aws_s3_bucket.payroll_docs: Refreshing state... [id=payroll-docs-eu-west-1-joao]
module.payroll.aws_instance.payroll_server: Refreshing state... [id=i-06ace3fd2f349e146]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # module.payroll_europe.aws_instance.payroll_server will be created
  + resource "aws_instance" "payroll_server" {
      + ami                                  = "ami-0324bce2436ce02b2"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t2.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = (known after apply)
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_group_id                   = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
      + private_dns                          = (known after apply)
      + private_ip                           = (known after apply)
      + public_dns                           = (known after apply)
      + public_ip                            = (known after apply)
      + region                               = "eu-west-1"
      + secondary_private_ips                = (known after apply)
      + security_groups                      = (known after apply)
      + source_dest_check                    = true
      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = (known after apply)
      + tags                                 = {
```

```
          + "Name" = "Payroll-Server"
        }
      + tags_all                           = {
          + "Name" = "Payroll-Server"
        }
      + tenancy                            = (known after apply)
      + user_data_base64                   = (known after apply)
      + user_data_replace_on_change        = false
      + vpc_security_group_ids             = (known after apply)

      + capacity_reservation_specification (known after apply)

      + cpu_options (known after apply)

      + ebs_block_device (known after apply)

      + enclave_options (known after apply)

      + ephemeral_block_device (known after apply)

      + instance_market_options (known after apply)

      + maintenance_options (known after apply)

      + metadata_options (known after apply)

      + network_interface (known after apply)
      + secondary_network_interface (known after apply)
    }
      + secondary_network_interface (known after apply)
    }

Plan: 1 to add, 0 to change, 0 to destroy.
module.payroll_europe.aws_instance.payroll_server: Creating...
module.payroll_europe.aws_instance.payroll_server: Still creating... [00m10s elapsed]
      + secondary_network_interface (known after apply)
    }

Plan: 1 to add, 0 to change, 0 to destroy.
module.payroll_europe.aws_instance.payroll_server: Creating...
module.payroll_europe.aws_instance.payroll_server: Still creating... [00m10s elapsed]
module.payroll_europe.aws_instance.payroll_server: Creation complete after 13s [id=i-0bd8d458e
f15cc5e6]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

Terraform Plan:

```
joaoribeiro@JOCAPC:~/remote-state-example/terraform$ terraform plan
module.payroll_europe.aws_dynamodb_table.payroll_db: Refreshing state... [id=PayrollDB-EU-joao
]
module.payroll.aws_dynamodb_table.payroll_db: Refreshing state... [id=PayrollDB-joao]
module.payroll.aws_s3_bucket.payroll_docs: Refreshing state... [id=payroll-docs-us-east-1-joao
]
module.payroll_europe.aws_s3_bucket.payroll_docs: Refreshing state... [id=payroll-docs-eu-west
-1-joao]
module.payroll_europe.aws_instance.payroll_server: Refreshing state... [id=i-0bd8d458ef15cc5e6
]
module.payroll.aws_instance.payroll_server: Refreshing state... [id=i-06ace3fd2f349e146]

Terraform used the selected providers to generate the following execution plan. Resource actio
ns are indicated with the following symbols:
  ~ update in-place

Terraform will perform the following actions:

  # module.payroll_europe.aws_instance.payroll_server will be updated in-place
  ~ resource "aws_instance" "payroll_server" {
        id                                   = "i-0bd8d458ef15cc5e6"
      ~ tags                                 = {
          ~ "Name" = "Payroll-Server-joao" -> "Payroll-Server"
        }
      ~ tags_all                             = {
          ~ "Name" = "Payroll-Server-joao" -> "Payroll-Server"
        }
        # (39 unchanged attributes hidden)

        # (9 unchanged blocks hidden)
    }

Plan: 0 to add, 1 to change, 0 to destroy.
```

Terraform init:

```
joaoribeiro@JOCAPC:~/remote-state-example/terraform$ terraform init
Initializing the backend...
Initializing modules...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.40.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

US Instance:

Instance summary for i-06ace3fd2f349e146 (Payroll-Server) Info

Updated less than a minute ago

Connect | Instance state | Actions

Instance ID i-06ace3fd2f349e146	**Public IPv4 address** 13.222.171.252 | open address	**Private IPv4 addresses** 172.31.25.252
IPv6 address –	**Instance state** Running	**Public DNS** ec2-13-222-171-252.compute-1.amazonaws.com | open address
Hostname type IP name: ip-172-31-25-252.ec2.internal	**Private IP DNS name (IPv4 only)** ip-172-31-25-252.ec2.internal	
Answer private resource DNS name –	**Instance type** t2.micro	**Elastic IP addresses** –
Auto-assigned IP address 13.222.171.252 [Public IP]	**VPC ID** vpc-0c191489f702038c7 (default)	**AWS Compute Optimizer finding** Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM role –	**Subnet ID** subnet-04e5e8b3ad0f851a7	**Auto Scaling Group name** –
IMDSv2 Optional EC2 recommends setting IMDSv2 to required | Learn more	**Instance ARN** arn:aws:ec2:us-east-1:686699774218:instance/i-06ace3fd2f349e146	**Managed** false
Operator –		

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ **Instance details** Info

AMI ID ami-07d9b9ddc6cd8dd30	**Monitoring** disabled	**Platform details** Linux/UNIX
AMI name ubuntu/images/hvm-ssd/ubuntu-jammy-22.04-amd64-server-20240207.1	**Allowed image** -	**Termination protection** Disabled
Stop protection Disabled	**Launch time** Tue Apr 14 2026 13:28:38 GMT+0100 (Hora de verão da Europa Ocidental) (25 minutes)	**AMI location** amazon/ubuntu/images/hvm-ssd/ubuntu-jammy-22.04-amd64-server-20240207.1
Instance reboot migration Default (On)	**Instance auto-recovery** Default	**Lifecycle** normal
Stop-hibernate behavior Disabled	**AMI Launch index** 0	**Key pair assigned at launch** –

EU Instance:

Instance summary for i-0bd8d458ef15cc5e6 (Payroll-Server-joao) Info

Updated 1 minute ago

Connect | Instance state | Actions

Instance ID i-0bd8d458ef15cc5e6	**Public IPv4 address** 34.245.95.41 | open address	**Private IPv4 addresses** 172.31.1.2
IPv6 address –	**Instance state** Running	**Public DNS** ec2-34-245-95-41.eu-west-1.compute.amazonaws.com | open address
Hostname type IP name: ip-172-31-1-2.eu-west-1.compute.internal	**Private IP DNS name (IPv4 only)** ip-172-31-1-2.eu-west-1.compute.internal	
Answer private resource DNS name –	**Instance type** t2.micro	**Elastic IP addresses** –
Auto-assigned IP address 34.245.95.41 [Public IP]	**VPC ID** vpc-0df92408ce1504f65	**AWS Compute Optimizer finding** Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM role –	**Subnet ID** subnet-0a0e833bff8558e2f	**Auto Scaling Group name** –
IMDSv2 Required	**Instance ARN** arn:aws:ec2:eu-west-1:686699774218:instance/i-0bd8d458ef15cc5e6	**Managed** false
Operator –		

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ **Instance details** Info

AMI ID ami-0324bce2436ce02b2	**Monitoring** disabled	**Platform details** Linux/UNIX
AMI name ubuntu/images/hvm-ssd-gp3/ubuntu-noble-24.04-amd64-server-20260313	**Allowed image** -	**Termination protection** Disabled
Stop protection Disabled	**Launch time** Tue Apr 14 2026 13:41:00 GMT+0100 (Hora de verão da Europa Ocidental) (12 minutes)	**AMI location** amazon/ubuntu/images/hvm-ssd-gp3/ubuntu-noble-24.04-amd64-server-20260313

S3 Buckets:

payroll-docs-eu-west-1-joao	Europe (Ireland) eu-west-1	April 14, 2026, 13:34:58 (UTC+01:00)
payroll-docs-us-east-1-joao	US East (N. Virginia) us-east-1	April 14, 2026, 13:28:38 (UTC+01:00)

DynamoDB Table:

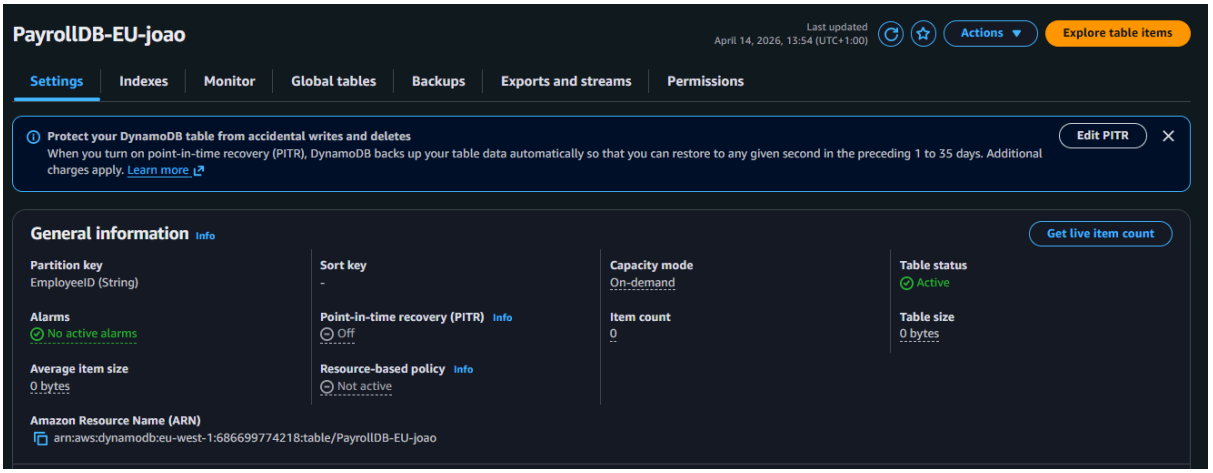
PayrollDB-EU-joao
Last updated
April 14, 2026, 13:54 (UTC+1:00)
Actions
Explore table items
Settings
Indexes
Monitor
Global tables
Backups
Exports and streams
Permissions
Protect your DynamoDB table from accidental writes and deletes
When you turn on point-in-time recovery (PITR), DynamoDB backs up your table data automatically so that you can restore to any given second in the preceding 1 to 35 days. Additional charges apply. Learn more
Edit PITR
General information Info
Get live item count
Partition key
EmployeeID (String)
Sort key
-
Capacity mode
On-demand
Table status
Active
Alarms
No active alarms
Point-in-time recovery (PITR) Info
Off
Item count
0
Table size
0 bytes
Average item size
0 bytes
Resource-based policy Info
Not active
Amazon Resource Name (ARN)
arn:aws:dynamodb:eu-west-1:686699774218:table/PayrollDB-EU-joao

Table: PayrollDB-EU-joao - Items returned (0)
Scan started on April 14, 2026, 13:55:11
Actions
Create item
1
EmployeeID (String)